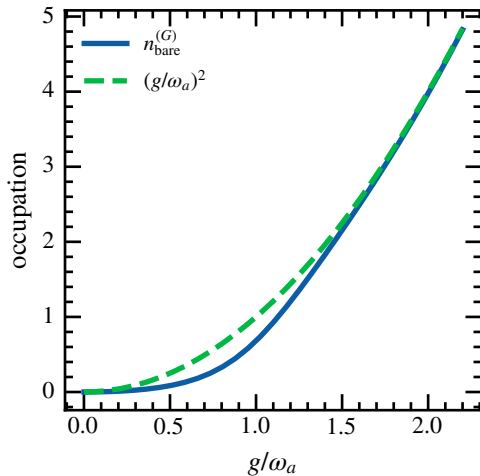
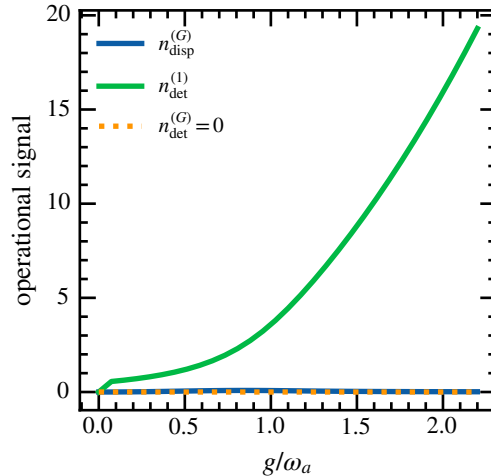
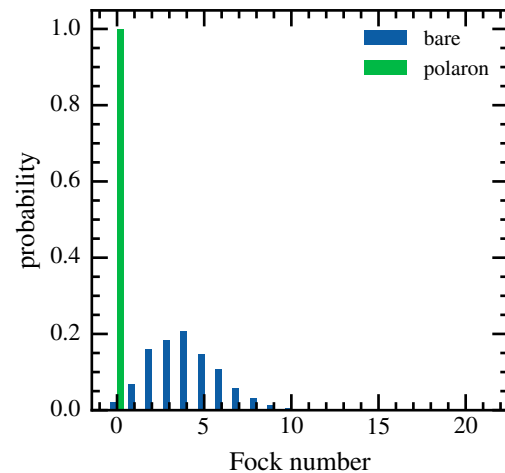


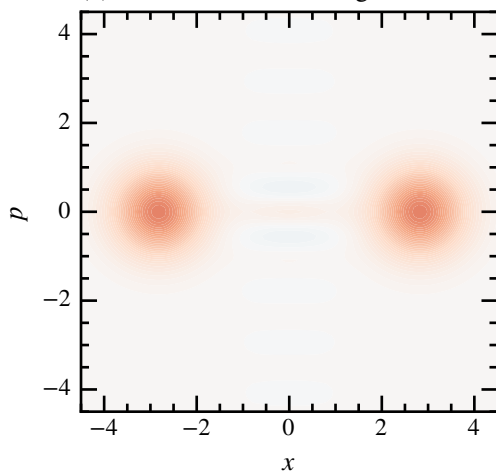
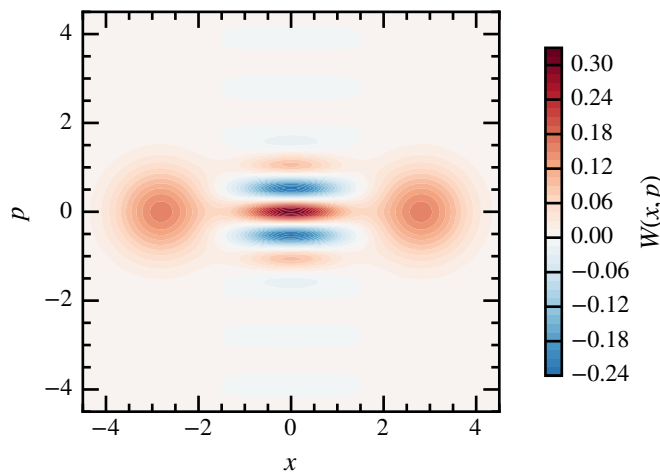
(a) Bare ground-state occupation



(b) Reference-dependent excitations

(c) $g/\omega_a=2$ 

(d) Reduced oscillator Wigner function

(e) Conditioned on $|g\rangle$ **Operational lesson**

Bare occupation measures dressing relative to the uncoupled vacuum.

Detector counts require the positive-frequency operator in the interacting basis.